

- Take the average information entropy of the current attribute.
- Calculate gain for the current attribute.
- 3. Pick the highest gain attribute.
- 4. Repeat until the desired tree is complete.

We will follow these steps while solving the numerical problem below.

Solved Problem 9.2

Consider the dataset given in Table 9.1. Construct the decision tree to predict whether a customer will buy a computer or not.

Solution:

Step 1: Compute Entropy

For computing the entropy, first we have to count the number of distinct outputs. In this case, decision has to be taken whether a customer will buy a computer or not. Out of the total 14 samples, based on the attributes like age, income, student, and credit rating, 9 samples belong to the "yes" category and 5 samples belong to the "no" category.

Number of "Yes" = 9

Number of "No" = 5

We have

$$\begin{aligned} \text{Entropy}(S) &\equiv -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus} \\ &= -\left(\frac{9}{14}\right) \log_2 \left(\frac{9}{14}\right) - \left(\frac{5}{14}\right) \log_2 \left(\frac{5}{14}\right) = 0.940 \end{aligned}$$

The initial entropy is computed using Eq. (9.1). For each of the attributes – age, income, student, and credit – the count of output class "buys_computer" belonging to the yes or the no class is first computed.

Step 2: Calculations for Every Attribute

In this case, the attributes are age, income, student, and credit. Based on whether a person buys a computer or not for the age group category, we can arrange the data as shown in Table 9.2.

Table 9.2 Data rearranged as per the attribute "age"

Attribute: AGE	Buy Computer	
	Yes	No
<30	2	3
31–40	4	0
>40	3	2
Gain	0.246	

We then carry out the calculations for each attribute. The entropy for each category of age (<30, 31–40, and >40) is computed. The gain of age is the difference between the overall entropy and the product of probability and entropy of individual instances. We have

$$\begin{aligned}\text{Gain}(S, A) &\equiv \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v) \\ \text{Gain}(S, \text{Age}) &= \text{Entropy}(S) - \left(\frac{5}{14}\right) * \text{Entropy}(S_{<30}) - \left(\frac{4}{14}\right) * \text{Entropy}(S_{31-40}) \\ &\quad - \left(\frac{5}{14}\right) * \text{Entropy}(S_{>40}) \\ &= 0.940 - \left(\frac{5}{14}\right) * 0.971 - \left(\frac{4}{14}\right) * 0 - \left(\frac{5}{14}\right) * 0.971 = 0.246\end{aligned}$$

Similarly, we calculate the gain of the other attributes.

For Income Attribute:

Based on the income of the person, we can arrange the data as shown in Table 9.3.

Table 9.3 Data rearranged as per the attribute "income"

<i>Attribute: INCOME</i>	<i>Buy Computer</i>	
	<i>Yes</i>	<i>No</i>
High	2	2
Medium	4	2
Low	3	1
Gain	0.029	

$$\begin{aligned}\text{Gain}(S, \text{Income}) &= \text{Entropy}(S) - \left(\frac{4}{14}\right) * \text{Entropy}(S_{\text{High}}) - \left(\frac{5}{14}\right) * \text{Entropy}(S_{\text{Med}}) \\ &\quad - \left(\frac{5}{14}\right) * \text{Entropy}(S_{\text{Low}}) \\ &= 0.029\end{aligned}$$

For Occupation Attribute:

Based on the occupation of the person, we can arrange the data as shown in Table 9.4.

Table 9.4 Data rearranged as per the attribute "occupation"

Attribute: STUDENT	Buy Computer	
	Yes	No
Yes	6	1
No	3	4
Gain	0.151	

$$\begin{aligned}\text{Gain}(S, \text{Student}) &= \text{Entropy}(S) - \left(\frac{7}{14}\right) * \text{Entropy}(S_{\text{Yes}}) - \left(\frac{7}{14}\right) * \text{Entropy}(S_{\text{No}}) \\ &= 0.151\end{aligned}$$

For Credit Attribute:

Based on the person's credit review, we can arrange the data as shown in Table 9.5.

Table 9.5 Data rearranged as per the attribute "credit"

Attribute: CREDIT	Buy Computer	
	Yes	No
Fair	6	2
Excellent	3	3
Gain	0.046	

$$\begin{aligned}\text{Gain}(S, \text{Credit}) &= \text{Entropy}(S) - \left(\frac{8}{14}\right) * \text{Entropy}(S_{\text{Fair}}) - \left(\frac{6}{14}\right) * \text{Entropy}(S_{\text{Excellent}}) \\ &= 0.046\end{aligned}$$

Step 3: Pick the Highest Gain Attribute

The attribute having the maximum gain is taken as the root attribute. Gain is a parameter that measures how much would be gained by branching on any attribute, say, A . It estimates the expected reduction in the information requirement caused by knowing the value of A . Thus, attribute A with the highest information gain is chosen as the splitting attribute at node N . This will result in the "best classification", so that the amount of information still required to terminate classifying the tuples is minimal.

In this case, $\text{gain}(S, \text{Age})$ is maximum; hence it is chosen as the splitting attribute. Using age as the root node, the initial split of attributes belonging to age <30 , age between 31 and 40, age >40 is arrived (Fig. 9.2). In the category age between 31 and 40, all the leaf nodes belong to the same category of $\text{buys_computer} = \text{yes}$ and hence it is a leaf node and not to be considered further.

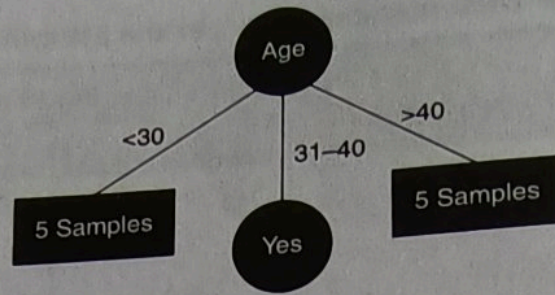


Figure 9.2 Age is chosen as the root node and the splitting of attributes are based on age.

Step 4: Repeat until Decision Tree is Complete

Recursively, we then find the splitting attributes at the next level. We now consider only part of the sample dataset for unclassified samples. Here, we have 5 samples for Age <30 and 5 samples for Age >40.

For Age <30:

The dataset for this category has 5 samples as given in Table 9.6.

Table 9.6 Dataset of persons whose age is <30

S. No.	Age	Income	Student	Credit	Buy
1	<30	High	No	Fair	No
2	<30	High	No	Excellent	No
8	<30	Medium	No	Fair	No
9	<30	Low	Yes	Fair	Yes
11	<30	Medium	Yes	Excellent	Yes

Based on whether a person buys computer or not and based on whether the buyer is student or not in the dataset, we construct Table 9.7.

Table 9.7 Dataset as per attribute "student" of people whose age is <30

Attribute: STUDENT	Buy Computer	
	Yes	No
Yes	2	—
No	—	3
Gain	0.971	

$$\begin{aligned}
 \text{Gain}(\text{Age} < 30, \text{Student}) &= \text{Entropy}(\text{Age} < 30) - \left(\frac{2}{5}\right) * \text{Entropy}(S_{\text{Yes}}) - \left(\frac{3}{5}\right) * \text{Entropy}(S_{\text{No}}) \\
 &= 0.971 - \left(\frac{2}{5}\right) * 0 - \left(\frac{3}{5}\right) * 0 \\
 &= 0.971
 \end{aligned}$$

Similarly, we can construct Table 9.8 based on the person's credit review.

Table 9.8 Dataset as per the attribute "credit" of people whose age is <30

<i>Attribute: CREDIT</i>	<i>Buy Computer</i>	
	<i>Yes</i>	<i>No</i>
Fair	1	2
Excellent	1	1
Gain	0.0208	

$$\begin{aligned}
 \text{Gain}(\text{Age} < 30, \text{Credit}) &= \text{Entropy}(\text{Age} < 30) - \left(\frac{3}{5}\right) * \text{Entropy}(S_{\text{Fair}}) - \left(\frac{2}{5}\right) * \text{Entropy}(S_{\text{Excellent}}) \\
 &= 0.971 - \left(\frac{3}{5}\right) * 0.917 - \left(\frac{2}{5}\right) * 1 = 0.0208
 \end{aligned}$$

Similarly, we can construct Table 9.9 based on the income.

Table 9.9 Dataset as per the attribute "income" of people whose age is <30

<i>Attribute: INCOME</i>	<i>Buy Computer</i>	
	<i>Yes</i>	<i>No</i>
High	0	2
Medium	1	1
Low	1	0
Gain	0.57	

$$\begin{aligned}
 \text{Gain}(\text{Age} < 30, \text{Income}) &= \text{Entropy}(\text{Age} < 30) - \left(\frac{2}{5}\right) * \text{Entropy}(S_{\text{High}}) \\
 &\quad - \left(\frac{2}{5}\right) * \text{Entropy}(S_{\text{Medium}}) - \left(\frac{1}{5}\right) * \text{Entropy}(S_{\text{Low}}) \\
 &= 0.971 - \left(\frac{2}{5}\right) * 0 - \left(\frac{2}{5}\right) * 1 - \left(\frac{1}{5}\right) * 0 = 0.57
 \end{aligned}$$

Since gain (Age <30, Student) is maximum, it is chosen as the splitting attribute.

For Age >40:

The dataset for this category has 5 samples as given in Table 9.10.

Table 9.10 Dataset of persons whose age is >40

S. No.	Age	Income	Student	Credit	Buy
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No
10	>40	Medium	Yes	Fair	Yes
14	>40	Medium	No	Excellent	No

Based on whether a person buys computer or not and based on the status of his/her credit in the dataset, we construct Table 9.11.

Table 9.11 Dataset as per attribute "credit" of people whose age is >40

Attribute: CREDIT	Buy Computer	
	Yes	No
Fair	3	0
Excellent	0	2
Gain	0.971	

$$\begin{aligned}
 \text{Gain}(\text{Age} > 40, \text{Credit}) &= \text{Entropy}(\text{Age} > 40) - \left(\frac{3}{5}\right) * \text{Entropy}(S_{\text{Fair}}) - \left(\frac{2}{5}\right) * \text{Entropy}(S_{\text{Excellent}}) \\
 &= 0.971 - \left(\frac{3}{5}\right) * 0 - \left(\frac{2}{5}\right) * 0 \\
 &= 0.971
 \end{aligned}$$

Similarly, we can construct Table 9.12 for income.

Table 9.12 Dataset as per attribute "income" of people whose age is >40

Attribute: INCOME	Buy Computer	
	Yes	No
High	0	0
Medium	2	1
Low	1	1
Gain	0.0208	

$$\begin{aligned} \text{Gain}(\text{Age} > 40, \text{Income}) &= \text{Entropy}(\text{Age} > 40) - \left(\frac{3}{5}\right) * \text{Entropy}(S_{\text{Medium}}) - \left(\frac{2}{5}\right) * \text{Entropy}(S_{\text{Low}}) \\ &= 0.971 - \left(\frac{3}{5}\right) * 0.917 - \left(\frac{2}{5}\right) * 1 = 0.0208 \end{aligned}$$

Since $\text{gain}(\text{Age} > 40, \text{Credit})$ is maximum, it is chosen as the splitting attribute. Thus, we can draw the final decision tree as shown in Fig. 9.3.

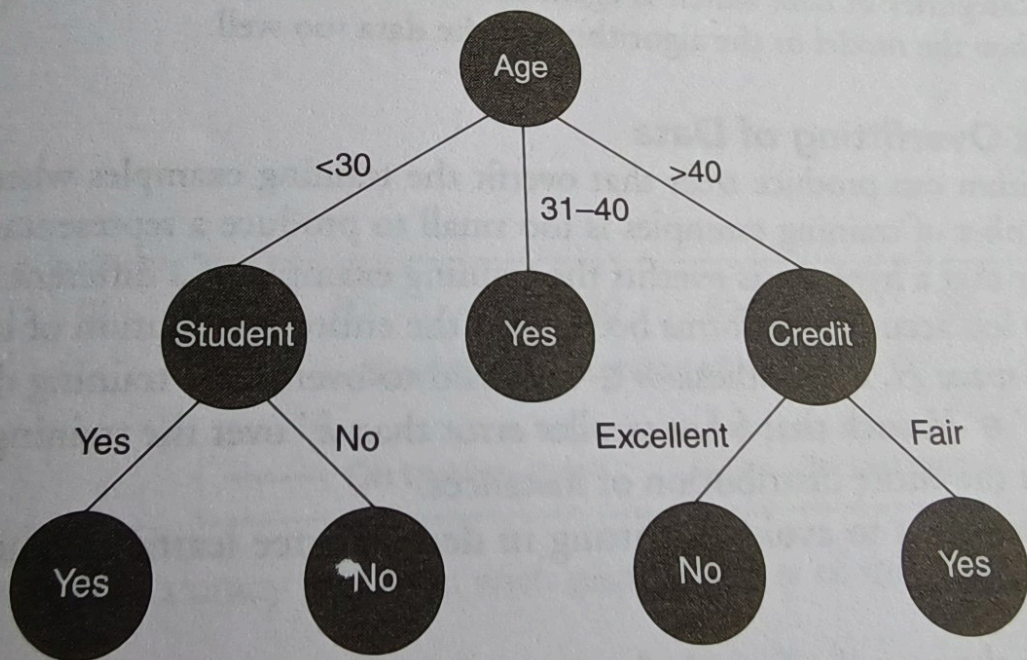


Figure 9.3 Final tree for buys_computer.